

11.29 (a) Following the same argument as in Problem 11.28 (a), we have the potentials in the reference frame K'

but expressed in the coordinates in K as $\Phi' = 0$ and $\vec{A}' = (\vec{m} \times \vec{R}) / R^3$, where $\vec{R} = \vec{r} - \vec{r}_0(t)$ and

$\vec{r}_0(t) = \vec{v}t$. By Lorentz transformation,

$$\Phi = \gamma(\Phi' + \vec{\beta} \cdot \vec{A}') = \gamma \frac{\vec{\beta} \cdot (\vec{m} \times \vec{R})}{R^3} = \gamma \frac{(\vec{\beta} \times \vec{m}) \cdot \vec{R}}{R^3}, \quad \vec{A}_{||} = \gamma(\vec{A}'_{||} + \vec{\beta} \Phi') = \gamma \vec{A}'_{||}, \quad \vec{A}_{\perp} = \vec{A}'_{\perp}.$$

Since $\gamma = 1 + O(\beta^2)$, then to the first order in β ,

$$\Phi = \frac{(\vec{\beta} \times \vec{m}) \cdot \vec{R}}{R^3}, \quad \vec{A}_{\perp} = \vec{A}'_{\perp}, \quad \vec{A}_{||} = \vec{A}'_{||}, \quad \text{or} \quad \vec{A} = \vec{A}' = \frac{\vec{m} \times \vec{R}}{R^3}.$$

(b) It is straightforward to verify that

$$\begin{aligned} \vec{E} &= -\nabla \Phi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t} = -\frac{\vec{\beta} \times \vec{m}}{R^3} + \frac{3[(\vec{\beta} \times \vec{m}) \cdot \vec{R}]\vec{R}}{R^5} + \frac{\vec{m} \times \vec{\beta}}{R^3} - \frac{3(\vec{m} \times \vec{R})(\vec{R} \cdot \vec{\beta})}{R^5} \\ &= \frac{3\vec{n}[\vec{n} \cdot (\vec{\beta} \times \vec{m})] - \vec{\beta} \times \vec{m}}{R^3} - \vec{m} \times \frac{3\vec{n}(\vec{n} \cdot \vec{\beta}) - \vec{\beta}}{R^3} = E_{\text{dipole}}(\vec{p}_{\text{eff}} = \vec{\beta} \times \vec{m}) - \vec{m} \times \frac{3\vec{n}(\vec{n} \cdot \vec{\beta}) - \vec{\beta}}{R^3} \end{aligned}$$

$$\text{Then, } \vec{E} = \frac{3\vec{n}[\vec{n} \cdot (\vec{\beta} \times \vec{m})/2] - (\vec{\beta} \times \vec{m})/2}{R^3} + \frac{3\vec{n}[\vec{n} \cdot (\vec{\beta} \times \vec{m})/2] - (\vec{\beta} \times \vec{m})/2}{R^3} + \frac{\vec{m} \times \vec{\beta}}{R^3} - \frac{3(\vec{m} \times \vec{n})(\vec{n} \cdot \vec{\beta})}{R^3}$$

$$= E_{\text{dipole}}(\vec{p}_{\text{eff}} = \frac{1}{2}\vec{\beta} \times \vec{m}) + \frac{3}{2} \frac{\vec{m} \times \vec{\beta}}{R^3} + \frac{3}{2} \frac{\vec{n}[\vec{n} \cdot (\vec{\beta} \times \vec{m})]}{R^3} + \frac{3(\vec{n} \times \vec{m})(\vec{n} \cdot \vec{\beta})}{R^3}$$

Notice that $\vec{n}[\vec{n} \cdot (\vec{\beta} \times \vec{m})] = \vec{n}[\vec{\beta} \cdot (\vec{m} \times \vec{n})] = \vec{\beta} \times [\vec{n} \times (\vec{m} \times \vec{n})] + (\vec{m} \times \vec{n})(\vec{n} \cdot \vec{\beta})$

$$= \vec{\beta} \times (\vec{m} - \vec{n}(\vec{m} \cdot \vec{n})) - (\vec{m} \times \vec{n})(\vec{n} \cdot \vec{\beta}) = -\vec{m} \times \vec{\beta} - (\vec{n} \times \vec{m})(\vec{n} \cdot \vec{\beta}) + (\vec{n} \times \vec{\beta})(\vec{n} \cdot \vec{m}), \text{ then}$$

$$\vec{E} = E_{\text{dipole}}(\vec{p}_{\text{eff}} = \frac{1}{2}\vec{\beta} \times \vec{m}) + \frac{3}{2} \vec{n} \times \frac{\vec{m}(\vec{n} \cdot \vec{\beta}) + \vec{\beta}(\vec{n} \cdot \vec{m})}{R^3}.$$

$$\text{Finally, } \vec{B} = \nabla \times \vec{A} = \frac{3\vec{n}(\vec{n} \cdot \vec{m}) - \vec{m}}{R^3}, \quad \text{and} \quad \vec{B} \times \vec{\beta} = \frac{3(\vec{n} \times \vec{\beta})(\vec{n} \cdot \vec{m}) - \vec{m} \times \vec{\beta}}{R^3}$$

Notice that $(\vec{n} \times \vec{\beta})(\vec{n} \cdot \vec{m}) = \vec{m} \times [(\vec{n} \times \vec{\beta}) \times \vec{n}] + \vec{n}[\vec{m} \cdot (\vec{n} \times \vec{\beta})] = \vec{m} \times (\vec{\beta} - \vec{n}(\vec{n} \cdot \vec{\beta})) + \vec{n}[\vec{n} \cdot (\vec{\beta} \times \vec{m})]$

$$= \vec{m} \times \vec{\beta} - (\vec{m} \times \vec{n})(\vec{n} \cdot \vec{\beta}) + \vec{n}[\vec{n} \cdot (\vec{\beta} \times \vec{m})], \text{ then}$$

$$\vec{B} \times \vec{\beta} = \frac{3(\vec{m} \times \vec{\beta} - (\vec{m} \times \vec{n})(\vec{n} \cdot \vec{\beta}) + \vec{n}[\vec{n} \cdot (\vec{\beta} \times \vec{m})]) - \vec{m} \times \vec{\beta}}{R^3} = \frac{3\vec{n}(\vec{n} \cdot (\vec{\beta} \times \vec{m})) - \vec{\beta} \times \vec{m}}{R^3}$$

$$- \frac{3(\vec{m} \times \vec{n})(\vec{n} \cdot \vec{\beta}) - \vec{m} \times \vec{\beta}}{R^3} = E_{\text{dipole}}(\vec{p}_{\text{eff}} = \vec{\beta} \times \vec{m}) - \vec{m} \times \frac{3\vec{n}(\vec{n} \cdot \vec{\beta}) - \vec{\beta}}{R^3} = \vec{E}.$$